

Math 141, S10
Exam 2

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	24	
2	24	
3	26	
4	26	
Total	100	

Good Luck !

Problem 1. (24 points) Find each requested derivative. Be sure to simplify your results.

(a) Find $f'(x)$ when $f(x) = \frac{1}{x^2} + \sqrt[3]{x^3}$

$$\begin{aligned} f'(x) &= \frac{d}{dx} [x^{-2} + x^{\frac{3}{2}}] \\ &= (-2)x^{-3} + \frac{3}{2}x^{\frac{1}{2}} \\ &= -\frac{2}{x^3} + \frac{3}{2}\sqrt{x} \end{aligned}$$

(b) Find $f'(x)$ when $f(x) = \sin x \cos x$

$$\begin{aligned} f'(x) &= \sin x (\cos x)' + \cos x (\sin x)' \\ &= -\sin^2 x + \cos^2 x \\ &= \cos 2x \end{aligned}$$

(c) Find $\frac{dy}{dx}$ when $y = \frac{x}{x^2 + 9}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x)'(x^2+9) - x \cdot (x^2+9)'}{(x^2+9)^2} \\ &= \frac{x^2+9 - x(2x)}{(x^2+9)^2} \\ &= \frac{-x^2+9}{(x^2+9)^2} \end{aligned}$$

(d) Find $\frac{dy}{dx}$ when $y = x^2 \ln x$

$$\begin{aligned} \frac{dy}{dx} &= x^2 \frac{d}{dx} [\ln x] + (\ln x) \frac{d}{dx} [x^2] \\ &= x^2 \cdot \frac{1}{x} + (\ln x) \cdot 2x \\ &= x + 2x \ln x \end{aligned}$$

Problem 2. (24 points) Find the derivative of each of the following functions.

(a) $f(x) = \left(\frac{x+1}{x-1}\right)^{10}$

$$\begin{aligned} f'(x) &= 10 \left(\frac{x+1}{x-1}\right)^9 \frac{d}{dx} \left[\frac{x+1}{x-1}\right] \\ &= 10 \left(\frac{x+1}{x-1}\right)^9 \frac{1 \cdot (x-1) - 1 \cdot (x+1)}{(x-1)^2} \\ &= 10 \left(\frac{x+1}{x-1}\right)^9 \cdot \frac{(-2)}{(x-1)^2} \\ &= -20 \frac{(x+1)^9}{(x-1)^{11}} \end{aligned}$$

(b) $f(x) = xe^{x-x^3}$

$$\begin{aligned} f'(x) &= x \frac{d}{dx} [e^{x-x^3}] + e^{x-x^3} \frac{d}{dx} [x] \\ &= x \cdot e^{x-x^3} \frac{d}{dx} [x-x^3] + e^{x-x^3} \\ &= x \cdot (1-3x^2) e^{x-x^3} + e^{x-x^3} \\ &= (x-3x^3+1) e^{x-x^3} \end{aligned}$$

(d) $f(x) = \log_{10} \sqrt{x-1}$

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[\frac{1}{2} \log_{10} (x-1) \right] \\ &= \frac{1}{2} \frac{1}{(x-1) \ln 10} \frac{d}{dx} [x-1] \\ &= \frac{1}{(x-1)(2 \ln 10)} \end{aligned}$$

(e) $f(x) = \tan^{-1}(x^2)$

$$\begin{aligned} f'(x) &= \frac{1}{1+(x^2)^2} \frac{d}{dx} [x^2] \\ &= \frac{2x}{1+x^4} \end{aligned}$$

Problem 3. (26 points)

(a) A super-ellipse is defined implicitly by the equation $16x^4 + y^4 = 32$. Use the **implicit differentiation** to find $\frac{dy}{dx}$, then find an equation for the tangent line to the super-ellipse at the point $(1, 2)$ based this result.

Solution: $\frac{d}{dx}[16x^4 + y^4] = \frac{d}{dx}[32]$

$$64x^3 + 4y^3 \frac{dy}{dx} = 0$$

$$4y^3 \frac{dy}{dx} = -64x^3$$

$$\frac{dy}{dx} = -16 \frac{x^3}{y^3}$$

At the point $(1, 2)$, we have

$$\frac{dy}{dx} = -16 \cdot \frac{1^3}{2^3} = -2$$

Thus, the equation of the tangent line is

$$(y - 2) = -2(x - 1)$$

$$y = -2x + 4$$

(b) Using the **logarithmic differentiation** to find the $\frac{dy}{dx}$ if $y = (x^2 + 1)^{\cos x}$.

Solution: $\ln y = \ln (x^2 + 1)^{\cos x}$

$$\ln y = \cos x \ln (x^2 + 1)$$

$$\frac{d}{dx}[\ln y] = \frac{d}{dx}[\cos x \ln (x^2 + 1)]$$

$$\frac{y'}{y} = \cos x \cdot \frac{1}{x^2 + 1} \cdot (2x) + (-\sin x) \ln (x^2 + 1)$$

$$y' = y \left[\frac{2x \cos x}{x^2 + 1} - (\sin x) \ln (x^2 + 1) \right]$$

$$= (x^2 + 1)^{\cos x} \left[\frac{2x \cos x}{x^2 + 1} - \sin x \ln (x^2 + 1) \right]$$

Problem 4. (26 points) Let $f(x) = x^4 - 8x^2 + 4$.

(a) Find all critical numbers of f , and determine the intervals on which f is *increasing* or *decreasing*.

$$f'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x+2)(x-2)$$

$$f'(x) = 0 \Rightarrow 4x(x+2)(x-2) = 0 \Rightarrow x = -2, 0, 2.$$

Intervals	$f'(x)$	
$(-\infty, -2)$	-	decreasing.
$(-2, 0)$	+	increasing.
$(0, 2)$	-	decreasing.
$(2, +\infty)$	+	increasing.

(b) Find the absolute maximum and absolute minimum values of f on the interval $[-3, 3]$.

$$f(0) = 4, \quad f(2) = -12, \quad f(-2) = -12$$

Two end points:

$$f(3) = 13, \quad f(-3) = 13$$

The absolute maximum value: 13

The absolute minimum value: -12

(c) Find the intervals on which f is *concave upward* or *concave downward*, and the inflection points.

$$f''(x) = 12x^2 - 16 = 0 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \frac{2}{\sqrt{3}} \text{ or } -\frac{2}{\sqrt{3}}.$$

Interval	$f''(x)$	
$(-\infty, -\frac{2}{\sqrt{3}})$	+	Concave upward
$(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$	-	Concave downward
$(\frac{2}{\sqrt{3}}, +\infty)$	+	Concave upward.

inflection points: $(\frac{2}{\sqrt{3}}, -\frac{44}{9})$

4 $(-\frac{2}{\sqrt{3}}, -\frac{44}{9})$

Extra Credits. (5 points) Find

$$\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^2(2x)}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^2(2x)} &= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 2x} \\ &= \lim_{x \rightarrow 0} \left(\frac{x}{\sin 2x} \right) \left(\frac{x}{\sin 2x} \right) \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$